MCO-1D COMMISSIONING MANUAL — REVISION 12

Addendum — Sections 53-58

This addendum covers all additions since Revision 11. Sections 1-52 remain as published in Revision 11. The wave library has grown to 41,779 lines across the session. Active biomimetic hypotheses: 20.

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SECTION 53 — REYNOLDS NUMBER AND MEDIUM SCALING

53.1 The Problem

Biological research on flapping locomotion spans organisms operating at Reynolds numbers from tens (tiny insects) to millions (large aquatic animals). The MCO-1D operates at Re 28,000 in hover rising to 89,000 in cruise. This is the hawkmoth and small bat regime. Mechanisms transfer across scales differently depending on whether they rely on dimensionless parameters or absolute values.

53.2 What Transfers Robustly

Dimensionless parameters transfer robustly across Reynolds number and across fluid medium. Phase offsets in degrees, amplitude ratios, Strouhal number, advance ratio, and stroke timing fractions are all geometry ratios — they do not depend on Re or medium density.

53.3 What Requires Calibration

Absolute AoA values	Must be calibrated to the S1223 polar at MCO-1D Re. S1223 CL_max at Re 50,000: approximately 18 degrees. Maximum L/D at approximately 7-8 degrees. Post-stall above approximately 22 degrees. Named constants: AOA_CL_MAX_DEG_RE50K = 18.0, AOA_BEST_LD_DEG_RE50K = 8.0.
LEV-dependent mechanisms	Clap-and-fling, reversal burst, sweep-clap. LEVs are proportionally smaller at MCO-1D Re than at insect Re. Mechanisms transfer but benefit magnitude is smaller. Hypotheses H-01, H-05, H-12, H-18 carry explicit Re notes.
Aquatic-source AoA values	Sea lion data (Drago 2025) gave thrust peak at 45 degrees AoA. This is valid for water with the sea lion foreflipper profile. On the S1223 in air at Re 50,000, 45 degrees is deep stall. H-19 corrected: thrust mode 45 degrees to 15 degrees, lift mode 82 degrees to 22 degrees.
Upstroke AoA floor	Sea turtle upstroke uses 2 degrees AoA in water where buoyancy provides partial weight support. In air there is no buoyancy. H-17 corrected: upstroke floor 2 degrees to 5 degrees.
H-04 pterosaur AoA	Original value 32 degrees from pterosaur-scale Re experiments at Re approximately 1,000,000. At MCO-1D Re 50,000 the S1223 stalls at approximately 18-22 degrees. Corrected to 18 degrees — S1223 CL_max at operating Re.

NOTE: The bat literature (H-02, H-22, H-23) operates at Re 10,000-100,000 — directly overlapping MCO-1D Re. These hypotheses require no Reynolds number correction.

SECTION 54 — BAT HYPOTHESES H-22 AND H-23

54.1 Why Bat Research is Most Applicable

Of all biological sources in the library, bat flight is the most directly applicable to the MCO-1D by Reynolds number. Bats operate at Re 10,000-100,000, directly overlapping the MCO-1D range of 28,000-89,000. No Reynolds number corrections are required. The 2025 paper by Fan and Breuer at Brown University, inspired by Cynopterus brachyotis (the lesser short-nosed fruit bat), provides the most recent quantified bat propulsion data directly applicable to ornithopter design at this scale.

54.2 H-22 — Bat Ventral Upstroke Clap

In the late upstroke, bat wings clap together below the body and produce an air jet that significantly increases lift production, with a positive peak matched to that produced in the downstroke. This creates a double lift peak — one from the downstroke and one from the upstroke clap — effectively two power strokes per wingbeat cycle without increasing frequency or amplitude.

Mechanism	At 85 percent through the upstroke, port and stbd wings of each pair sweep inward, dihedral angles toward the fuselage. Then rapidly fling apart at 1.3 times normal sweep speed. The fling produces a downward jet generating upward thrust.
Distinct from H-18	H-18 (sea turtle) is a sweep-clap at the BOTTOM OF DOWNSTROKE. H-22 (bat) is a ventral clap at the BOTTOM OF UPSTROKE. Different timing, different axis, complementary mechanisms.
Power economy	Lift and power economy increase with wing folding. Economy peaks at MODERATE clap angle — not maximum. Four variants: 10, 20, 30, and 35 degrees. Moderate (20 degrees) expected to give best lift per watt.
Re note	No correction needed. Direct Re overlap with MCO-1D. Source: PMC11837331 (2025) — Fan, Breuer et al., Brown University.

54.3 H-23 — Bat Speed-Adaptive Upstroke AoA

The function of the upstroke changes with forward flight speed in bats. At hovering and slow speed, bats use an inverted wing during the upstroke to generate additional lift. At high speed, the upstroke is feathered or slightly negative, producing a reverse vortex loop that reduces induced drag on the following downstroke.

Hover (V below 2 m/s)	Inverted upstroke: AoA -8 degrees. Both strokes generate lift.
Transition (2-6 m/s)	Smooth ramp from -8 to +5 degrees AoA.
Cruise (6-9 m/s)	Feathered: AoA +5 degrees. Minimal drag.
Sprint (above 9 m/s)	Slight negative: AoA blends to -3 degrees. Reverse vortex loop reduces drag on next downstroke.

NOTE: H-23 is speed-adaptive in a fundamentally different way from H-17 (sea turtle asymmetric stroke). H-17 uses a fixed upstroke profile regardless of speed. H-23 changes sign with speed — inverted at hover, positive at cruise,

negative at sprint.

SECTION 55 — WATER OPERATIONS SOFTWARE FRAMEWORK

55.1 Philosophy — Not a Seaplane

The MCO-1D should not be operated as a conventional seaplane. Seaplane procedures — porpoising on water runs, climbing over the hump, spray management at planing speed — are irrelevant to an aircraft that departs and arrives vertically. The MCO-1D has fundamental advantages over any conventional seaplane: a stall speed of 4.85 m/s (lower than any fixed-wing waterborne aircraft), vertical departure capability that bypasses the hump entirely, and hover capability that makes glassy water landings trivial. The software framework supports what the MCO-1D actually needs.

55.2 Wave Height — FAA Practical Measurement

The system uses FAA-standard practical wave height measurement: crest to trough in feet and inches, as observed by the pilot or estimated from the rangefinder time series. Significant wave height (Hs) and sea state numbers are not used — they are not the standard for seaplane operations and are not available to pilots.

Calm / glassy	3 inches (0.08m) — glassy water approach protocol
Light seaplane safe	15 inches (0.38m) — FAA AC 150/5395-1B
Restriction begins	18 inches (0.46m) — normal ops restricted
Expert pilots only	20 percent of float length (float-dependent)
Do not attempt	50 percent of float height keel to deck

55.3 Wave-Timed Departure — FAA Doctrine

FAA-H-8083-23: Open the throttle to takeoff power just as the floats begin RISING on a wave. The float rides up over the crest, already on the step when it arrives. For MCO-1D vertical departure: initiate the jump as the float begins to rise, so liftoff occurs at or near the crest. The crest is the highest point of the wave — minimum climb required to clear the water surface.

> WARNING: Departing from the wave trough is wrong. The aircraft would be at its lowest point with an oncoming wave face directly ahead.

55.4 Porpoising Procedure — FAA H-8083-23

If porpoising is detected (divergent pitch oscillation on water): First oscillation: apply and maintain back elevator pressure. Do not chase the oscillations. If not stopped by the second oscillation: reduce power to idle, hold full elevator back, allow the aircraft to settle onto water.

> NOTE: Water operations hardware — floats, immersion sensors, float inflation system — are deferred to a future BOM update when float geometry is specified. Hover capability must be demonstrated at TC-02 before any water operations are planned.

SECTION 56 — PRECISION DESCENT RATE CONTROLLER

56.1 The FAA Standard

FAA-H-8083-23 on glassy water landings: establish a stabilized approach and rate of descent of approximately 100 to 150 fpm using power and pitch changes. This is the validated target for precision water approach. It applies equally to IFR approaches over any surface where visual height reference is absent.

56.2 Physics — Is 100 fpm Achievable?

The physics strongly suggests yes. At estimated hover (2.5 Hz), the difference between hovering and descending at 100 fpm is approximately 3.6 percent of hover power — a frequency change of approximately 0.045 Hz. The wave library controls frequency to 0.01 Hz resolution. The AoA change required is approximately 1.2 degrees at forward speed, well within the servo resolution of 0.1 degrees.

> CAUTION: These are physics-based estimates. Actual hover frequency, lift-frequency curve shape, and bandwidth are unknown until TC-02. The controller architecture is complete; gain constants require calibration at TC-02 via calibrate_descent_controller().

56.3 Control Architecture

Height source	Rangefinder primary below 10m — measures directly to water surface. Barometric secondary and cross-check. Display in feet (pilot units) — not metres.
Inner loop: frequency	Proportional to descent rate error. Descending too fast: increase frequency (more lift, slow descent). Too slow or hovering: decrease frequency (less lift, begin descent). Fastest axis — responds within 1-3 wingbeat cycles.
Secondary loop: AoA	Fine trim that maintains the target rate against slow drift. Maximum correction: 3 degrees AoA change.
Outer loop: height integral	Slow integration prevents accumulated small errors from causing the descent rate to gradually diverge.
Target rates	100 fpm (0.508 m/s): IFR water approach, FAA standard. 300 fpm (1.524 m/s): normal instrument approach. 500 fpm (2.540 m/s): rapid non-precision descent.
Commands	descent_ctrl_engage 100 — engage at 100 fpm. descent_ctrl_disengage — return to normal. descent_ctrl_status — show current state.

> NOTE: The glassy water approach protocol applies the 100 fpm standard and instructs the pilot: do not judge altitude visually, trust the rangefinder only, continue descent until touchdown without flattening out. This follows FAA H-8083-23 exactly.

SECTION 57 — PRECISION SCORING AND DEMANDING INPUT RULE

57.1 What Precision Measures

Precision is measured from four rolling sigma values computed every tick during stable flight. Each sigma is the standard deviation of a controlled variable over a three-second window. Lower sigma means tighter control — the aircraft drifts less from its commanded condition. The four axes are altitude, roll angle, heading, and descent rate. Together they describe how smoothly the aircraft holds whatever condition the pilot has set.

Sigma h	Altitude deviation. Primary for water approach and SAR loiter.
Sigma roll	Roll angle deviation. Primary for transit heading hold.
Sigma hdg	Heading deviation. Primary for transit.
Sigma sink	Descent rate deviation. Primary when descent controller is active.

57.2 The Demanding Input Rule

When the pilot makes a demand, precision is irrelevant. The aircraft should give everything it has. Precision only matters when the pilot is holding steady. Three triggers force performance-only scoring:

Full stick input	Any stick above 70 percent deflection. Full roll, full yaw, or full pitch: performance only.
Lever at extremes	Lever above 88 percent (full power) or below 12 percent (maximum airbrake). Pilot commanding maximum authority: give it.
Abrupt control movement	Lever rate above 200 percent per second. Sudden input: performance only until input stabilises.

When any trigger fires, the phase objective score reverts to raw Ps and the precision history is cleared. When the pilot re-steadies the controls, the precision history rebuilds from zero over three seconds.

> NOTE: This is the same principle a good aerobatic aircraft should have: crisp and responsive to commands, stable and precise when held. The two qualities do not conflict — they apply at different times.

SECTION 58 — ACTIVE DAMPING AND PRECISION AEROBATICS

58.1 The Aerobatic Precision Problem

Precision aerobatics requires two qualities that operate at different timescales. The start and finish of each element should be crisp — the control input must be decisive and fast. The rhythm and rate of roll must be consistent throughout — once established at the hesitation angle, it must hold precisely. A conventional aircraft cannot change its aerodynamic damping characteristics mid-manoeuvre. The MCO-1D can.

58.2 Active Damping — How It Works

When the roll stick returns to neutral with residual angular velocity, the active damping system detects the residual roll rate and immediately applies opposing differential amplitude. In the verification test, a 45 degree per second residual rate was opposed with a 34-degree amplitude differential: port wings at 62 degrees, starboard at 96 degrees. This is applied every 20 milliseconds until the roll rate drops below the settled threshold of 2 degrees per second.

ROLLING (stick deflected)	Full performance. Demanding input rule applies. Maximum differential amplitude for commanded roll rate. No damping applied.
DAMPING (stick just neutral)	Opposing differential amplitude applied at 50 Hz. Target attitude frozen at the angle where stick went neutral. Correction proportional to residual roll rate.
HOLDING (rate settled)	Symmetric wing configuration. Precision sigma_roll measured during this phase. Attitude held without active correction — damping complete.

58.3 The Eight-Point Hesitation Roll

For an eight-point hesitation roll, the pilot applies roll stick, holds for 45 degrees of rotation, releases stick, aircraft damps and holds, applies stick again for the next 45 degrees, releases, aircraft damps and holds. The rhythm and rate of roll must be consistent throughout and between any hesitation elements. The crispness of each hesitation is measured as the time from stick-neutral to roll rate below 2 degrees per second.

58.4 Aerobatic Score

The aerobatic score combines crispness and stability into a single metric. Crispness is the mean settle time across recent manoeuvres. Stability is the standard deviation of roll angle during hold phases. A crisp entry with a wobbly hold scores poorly. A clean hold with a mushy entry also scores poorly. The aerobatic focus mode in Discovery Mode runs exploration specifically to find configurations that score well on both.

> WARNING: All active damping gain constants are marked with the note CALIBRATE TC-02. Actual roll rates, angular momentum, and required differential amplitudes are unknown until first flight. The architecture is complete; the gains require calibration from measured TC-02 data via the calibrate function.

SECTION 59 — UPDATED STATISTICS — REVISION 12

59.1 Software Statistics

Wave Library	41,779 lines
Total Python files	10 files, ~43,000 lines total
Active biomimetic hypotheses	20 (H-01 to H-23, excluding H-09/10/11)
Hypothesis variants	approximately 80 variants across all hypotheses
Biological sources	11 animals spanning insects to marine mammals
Re overlap with MCO-1D	Best: bats (H-02, H-22, H-23) — direct Re match
Discovery focus modes	10 (full, gust, fault, cruise, hover, envelope, biomimetic, precision, water_operations, aerobatic)
Precision descent targets	100 fpm, 300 fpm, 500 fpm
Active damping	Full: detect, oppose, damp, hold
TC-02 calibrations needed	Descent controller, active damping gains

59.2 Key Pending Items — Revision 12

TC-02 first priority	Measure hover frequency, lift-frequency curve slope, roll rate response, angular momentum characteristics. These calibrate: descent controller, active damping, and confirm all Re-corrected AoA starting values.
Float specification	Water operations wave height limits, capsize stability, and operating procedures cannot be finalised until float geometry is specified.
Hover confirmation	Water operations, precision descent approach, and active damping all assume the aircraft can hover. TC-02 confirms or refutes this.
H-09, H-10, H-11	Still pending TC-02 ODrive validation and bench testing.
GPS position controller	Discovery Mode racetrack precision improved with GPS hold. Currently heading-plus-time only.
mco_loadcell.py	HX711 load cell reader for TC-02 static thrust measurement.

59.3 The 20-Hypothesis Biological Library

H-01 Dragonfly	Phase switching — in-phase for force

H-02 Bat (advance)	Advance rotation — pre-stroke AoA transition
H-03 Manta	Spanwise wave — phase gradient root to tip
H-04 Pterosaur	Slow soar — 0.8Hz, 18 degrees AoA (Re-corrected)
H-05 Hawkmoth	LEV management — spanwise amplitude gradient
H-06 Negative cascade	W4 leads W1 — reversed cascade direction
H-07 Split frequency	10 percent inter-pair differential
H-08 Strouhal	Target St 0.20-0.40
H-12 Locust	Reversal burst — ODrive burst at stroke reversal
H-13 Plesiosaur phase	Speed-adaptive cascade phase
H-14 Plesiosaur brake	Anti-augmentation braking
H-15 Plesiosaur dominant	Pair amplitude dominance switching
H-16 Sea turtle figure-8	Dynamic in-stroke sweep — new kinematic
H-17 Sea turtle asymmetric	Power downstroke, recovery upstroke
H-18 Sea turtle clap	Sweep-clap at bottom of downstroke
H-19 Sea lion AoA	15 degrees thrust, 22 degrees lift (Re-corrected)
H-20 Otter advance ratio	Drag-to-lift mode switching with speed
H-21 Sea lion burst	High-velocity clap initiation fling
H-22 Bat ventral clap	Double lift peak — upstroke ventral clap
H-23 Bat upstroke AoA	Speed-adaptive — inverted to feathered to negative